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Course Name: Data structure

Course code: CSA 0310

Q:1 Find the bugs , including the missing lines in the following snippet

Your Fixed Code

```
1 #include<stdio.h>
2 int linearsearch(int arr[],int n,int key){
3     for(int i=0;i<n;i++){
4         if(arr[i]==key){
5             return i;
6         }
7     }
8     return -1;
9 }
10 int main(){
11     int arr[]={10,20,30,40,50};
12     int key=30;
13     int result=linearsearch(arr,5,key);
14
15     if(result!=-1)
16         printf("found at index%d",result);
17     else
18         printf("Not found");
19     return 0;
20 }
```

✓ Submitted

Test Input

30

Output

found at index2

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Q:2 Buggy code in an array

The screenshot shows a coding platform interface with the following components:

- QUESTIONS**: 2 / 8 completed. A dropdown menu shows "Character".
- CHARACTER**: A description: "Write a program that takes a character as input and determines whether it is a vowel or a consonant."
- PROGRAM EDITOR**: Contains the following C code:

```
1 #include <stdio.h>
2 int main()
3 {
4     char ch;
5     scanf("%c", &ch);
6     if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' || ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U')
7         printf("Vowels");
8     else
9         printf("Consonant");
10    return 0;
11 }
```
- INPUT**: An empty text box.
- OUTPUT**: An empty text box.
- Run** and **Save** buttons are present above the editor.
- Snipping Tool**: A floating window in the bottom right corner with the message: "Screenshot copied to clipboard. Automatically saved to screenshots folder. Mark-up and share".

Q:3 Identify an errors in the following snippet

```
1 #include<stdio.h>
2 int binarysearch(int arr[],int n,int key){
3     int low=0,high=n-1;
4     while(low<high){
5         int mid=(low+high)/2;
6         if (arr[mid]==key)
7             return mid;
8         else if(arr[mid]<key)
9             low=mid+1;
10        else
11            high=mid-1;
12    }
13    return -1;
14 }
15 int main(){
16     int arr[]={10,20,30,40,50,6,70,};
17     int n=7;
18     int key;
19
20     printf("Enter the element to search");
21     scanf("%d",&key);
22
23     int result=binarysearch(arr,n,key);
24
25     if(result!=-1)
26         printf("element found at index%d\n",result);
27     else
28         printf("element not found\n");
```

40

Output

```
Enter the element to searchelement
found at index3
```

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